

Machine learning For DDoS attacks Detection

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Abstract

Distributed Denial of Service (DDoS) attacks are becoming more complex and frequent happening, highlighting the need for advanced methods to detect them and safeguard modern networks. This project focuses on utilizing supervised machine learning to accurately and efficiently identify DDoS attacks, addressing the limitations of traditional detection methods. This study uses the CICID2017 dataset and compares four machine learning algorithms Random Forest (RF), XGBoost, Logistic Regression (LR), and Naive Bayes (NB). Random forest feature selection was used to improve the model performance and reduce computational overhead.

The results show that both XGBoost and Random Forest performed extremely well, achieving perfect accuracy, precision, and AUC (Area Under the Curve) scores of 1.0 across different sets of features (all feature set, 53, 13, 8), while also keeping false alarm rates very low (below 0.05%). In contrast, Logistic Regression and Naive Bayes showed reduced performance when fewer features were used, highlighting the importance of choosing the right features. To confirm the models' reliability, stratified 5-fold cross-validation was used, showing average accuracies of 99.99% for both RF and XGBoost. A sanity check was also conducted by shuffling the labels, which reduced the accuracy to about 50–56%, proving that the models learned from actual patterns and were not overfitting or affected by data leakage.

Among the models, XGBoost stood out for its speed and high accuracy, taking only 0.06 minutes to train on the full dataset. Feature selection using Random Forest Feature Importance (RFFI) was effective, as models with only 8 features still achieved 99.93% accuracy using XGBoost. This suggests that the original dataset had many unnecessary features. Based on these findings, XGBoost and other ensemble models are recommended for systems that need high accuracy, while simpler models like Logistic Regression may be used when many features are available.

This study provides useful guidance for applying machine learning to intrusion detection systems (IDS). Future work should explore detection of multiple types of attacks, improve resistance to adversarial attacks, and develop real-time detection systems. The results support the use of machine learning to tackle the growing DDoS threats and highlight the need for solutions that are not only accurate but also understandable and adaptable to different environments.

Keywords: DDoS, machine learning, Feature selection, cybersecurity, CICIDS2017, Machine learning, Random Forest, Logistic Regression, XGBoost, Naive Bayes, Stratified cross validation.

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Chapter 1: Introduction

Today, the world is interconnected through the internet, and network security has become one of the critical concerns for both individuals and organisations that rely heavily on digital networks for daily business operations such as communication, information sharing, and commerce (msspcolumbus, 2023; Barney and Lutkevich, 2022). The rise of advanced technologies and resources has increased the complexity and frequency of cyber threats, with Distributed Denial of Service (DDoS) attacks being one of the most prominent examples (McCann, 2024).

A Distributed Denial of Service (DDoS) attack occurs when attackers take control of multiple compromised devices, known as zombies, and use them to flood a target system with excessive traffic. This includes using overwhelming flow of messages, data packets, or service requests making the targeted system or service slow or entirely unable to legitimate users (Cloudflare, 2024).

DDoS attacks are generally classified into three types: volume-based, protocol-based, and application-based, each targeting different layers of the OSI model (Fastly, 2021). Volume-based attacks targets Layer 3 by overwhelming the network with excessive traffic, consuming bandwidth, and using botnets to render unresponsive systems (Imperva, 2021). Whereas Protocol-based attacks exploit vulnerabilities in Layer 4, such as SYN floods, which disrupt the TCP handshake and prevent legitimate users from establishing connections. Unlike volume based and protocol based, Application-based attacks focus on Layer 7, exploiting weaknesses in web services to cause disruptions. A well-known example is the Slowloris attack, which overwhelms servers by keeping connections open with incomplete HTTP requests, effectively blocking access for legitimate users (Shorey et al., 2018).

These attacks have evolved significantly over the years. The concept of Denial of Service (DoS) attacks started back to 1974 when a student disrupted the PLATO terminal at the University of Illinois. Since then, DDoS attacks have become increasingly sophisticated and widespread (Radware and Radware, 2017). According to Microsoft 365 (2023), these attacks are occurring more frequently and on a larger scale, with projections indicating further escalation. Modern DDoS attacks can reach peak volumes of multiple terabits per second, as demonstrated by incidents targeting major platforms such as GitHub, AWS, and Microsoft Azure (Microsoft 365, 2023). These attacks overwhelm systems, leading to downtime, financial losses, and reputational damage for organizations.

The increasing frequency and intensity of these attacks highlight the critical need for early detection, mitigation, and prevention. Traditional detection methods, such as signature-based and threshold-based approaches, are ineffective in identifying complex or zero-day attacks, as they rely on static rules or predefined patterns. This limitation makes them incapable of detecting evolving attack behaviours and previously unseen threats (Mittal, Kumar and Behal, 2023). Additionally, they struggle to differentiate between malicious and legitimate traffic, leading to false positives or undetected breaches (Wei *et al.*, 2021). Due to these limitations, traditional methods alone cannot effectively defend against modern DDoS attacks. As a result, more adaptive and intelligent solutions are required such as machine learning.

Machine learning offers a dynamic and scalable approach to detecting DDoS attacks by analyzing network traffic in real-time. These models can handle large, dynamic datasets and identify abnormal traffic patterns with greater accuracy (Mittal et al., 2023; Aslam, Srivastava and Gore, 2024). Also, their success in intrusion detection systems (IDS) makes them a promising solution for real-time DDoS attack detection. Moreover, due to their adaptability to changing attack patterns and learning from labelled data, machine learning enhances the precision and responsiveness of network security systems.

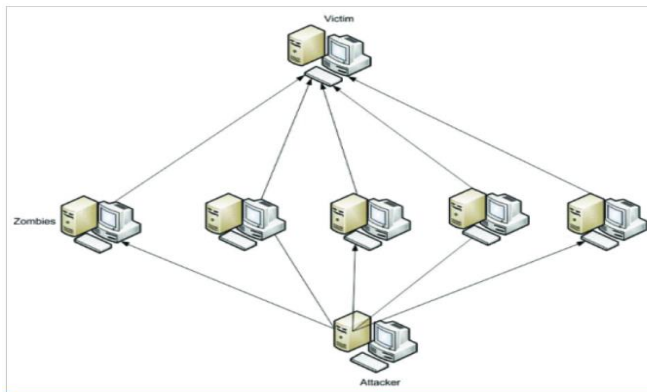


Figure 1: DDoS attack architecture.

Aims and Scope of the project

The aim of this project is to design, develop and evaluate a supervised machine learning model for the accurate and efficient detection of DDoS attacks using the feature selection in a network traffics. Using the label dataset such as CICIDS 2017, this model will focus on classify network traffic as normal traffic and DDoS attack, improving the accuracy whilst checking speed of the intrusion detection system (IDS) for real time use.

Objectives

1. To conduct a study on the working mechanism of DDoS attack and challenges for detection.
2. Exploring the current Machine learning detection models and choosing the best model for this project.
3. Using the CICIDS 2017 Dataset for Training and validating the selected machine learning, ensuring the proper application of preprocessing and feature selection technique
4. Using the random forest feature importance as the feature selection and using selected feature for training the other models.
5. Evaluating and comparison the popular models such as random forest, logistic regression etc. using the (accuracy, precision, recall, F1-score)
6. Use Receiver Operating Characteristics (ROC) CURVES AND Area Under Curve (AUC) metric for comparing the model performance.

Report Structure

The structure of the report is classified as follows:

Chapter1: Introduction

This chapter highlights basic understanding of DDoS, how it works and its impacts in today's world. Why machine learning is important.

Chapter 2: Literature review

This chapter is for finding related work and giving a critical review on relevant work, including existing DDoS detection methods and machine learning techniques in cybersecurity.

Chapter 3: Requirements

This chapter includes system requirements necessary for successful development of the DDoS attack detection model using machine learning.

Chapter 4: Methodology

This chapter is for describing the methodology of the project such as workflow of machine learning pipeline, including dataset selection, data preprocessing, model training, and evaluation techniques.

Chapter 5: Result and Discussion

This chapter presents the results of the experiments, including performance evaluation and comparison of different models.

Chapter 6: Project Evaluation

This chapter will report on a summary of findings and recommendations for future research.

Chapter 7: Further Work and Conclusions

This chapter is for reflection on the work such as planning and management, strength, and weakness to further development.

Chapter 2: Literature review

2.1 Introduction to Machine Learning for DDoS Detection

Machine learning for DDoS detection has been widely studied due to its adaptability, automation, and accuracy on detection. According to Sommer and Paxson machine learning-based intrusion detection systems can analyze the traffic and identify attack patterns without requiring detailed human instruction. Their ability to continuously learn from new data, allowing them to adjust the evolving threats and recognize new types of attacks. This adaptability of system is especially crucial in today's rapidly changing threat landscape. As it can detect unusual behaviours more effectively, making them a more robust solution for DDoS prevention.

Among the several types of machine learning, supervised and unsupervised learning are the most used in DDoS detection systems. Supervised learning relies on labeled datasets to distinguish between normal and malicious traffic. Algorithms such as k-Nearest Neighbors (KNN), Decision Trees (DT), Support Vector Machines (SVM), and Naïve Bayes have been widely adopted for this purpose (Mirjalili, Faris and Aljarah, 2020). These models learn from historical attack data to identify patterns, predict behaviors, and classify traffic in real-time.

On the other hand, unsupervised learning operates on unlabeled data, detecting anomalies by identifying unusual or irregular patterns in network traffic. Clustering techniques like K-Means and association rules have been effectively used for this task (Wang et al., 2020).

While supervised models offer high accuracy when trained on reliable datasets, they may struggle to detect unknown threats. Meanwhile, unsupervised methods are more flexible in identifying new attack patterns but often face challenges with false positives. The choice between these approaches depends on factors such as dataset availability and the intended application, whether for real-time detection systems, educational research, corporate security, or other use cases.

2.2 Related work on different Environments

The application of machine learning in DDoS has expanded across many environments, including Software-Defined Networking (SDN), Internet of Things (IoT), and Cyber-Physical Production Systems (CPPS), each posing unique challenges.

In the SDN environment, Alhamami and Albermany (2023) evaluated the effectiveness of different models using the Mininet emulator and OpenDaylight controller, with XGBoost and Logistic Regression (LR) other models achieving detection accuracies 99.26% and 99.24% respectively. However, their reliance on simulated datasets reduces the generalizability of the results to real-world deployments. To address adaptability, Alashhab et al. (2024) proposed an ensemble online learning framework for SDN, integrating Bernoulli Naïve Bayes (BernoulliNB), Passive-Aggressive, and Stochastic Gradient Descent (SGD) classifiers. This model attained a 99.2% detection rate on diverse datasets, including CICDDoS2019, InSDN, and slow-read-DDoS. Nevertheless, the complexity involved in maintaining and updating the ensemble model poses challenges for continuous operation.

In IoT networks, Ullah et al. (2023) demonstrated the effectiveness of Random Forest (RF) and Decision Tree (DT) classifiers on the CICI-IDS-2018 dataset, achieving a 99.98% detection rate with

low computational overhead. Yet, the use of under sampling to address class imbalance may compromise the model's generalizability to unseen data.

In the CPPS context, Hussain et al. (2024) developed a hybrid intrusion detection system that combined rule-based mechanisms with supervised ML models. Among them, XGBoost outperformed unsupervised techniques in detecting DDoS threats. However, the system's dependence on predefined rules limits its adaptability across different industrial settings and evolving threat scenarios. These studies collectively highlight the growing relevance and effectiveness of ML techniques in securing diverse network infrastructures.

2.3 Commonly used Supervised learning

Supervised Machine learning algorithms such as Random Forest, XGboost and logistic Regression have consistently shown a strong performance in identifying DDoS attacks in terms of accuracy. As reported by Alhamami and Albermany (2023), XGBoost (99.26%) and Logistic Regression (99.24%) within a Software-Defined Network (SDN) simulation. Similarly, Najjar and Naik (2022) found RF achieving a 99.13% detection rate. Similarly, Farhat et al. (2023) evaluated multiple ML models on the CICIDS2017 dataset and found that XGBoost achieved 99.11% accuracy with a remarkably low false alarm rate of 0.011%, significantly surpassing traditional algorithms like Logistic Regression and Support Vector Machines.

Al-Eryani et al. (2023) further validated the efficacy of Decision Trees and Logistic Regression, noting their efficient training time and suitability for high-speed environments. This robustness of algorithms is because of their ability to handle high dimensional data and their resistance to noise, which are critical factors in real-world network environment where traffic patterns can be highly variable (Jiyad et al., 2024).

Despite the accuracy and performance, RF and XGBoost requires significant processing power and resources, making them unsuitable for environment like Internet of Things (IoT) Ullah et al., (2023). Also, their black box in nature creates the trust issues for security analyst to trusting the decision and these types of model work best with clean and structured data making it less reliable on real world settings where data is constantly changing (Jiyad et al., 2024).

2.4 Adoption of Ensemble and Online Learning for Adaptability

To address this issues, Ensemble and online learning methods are begin explored for adapting to evolving attack patterns and reduce false positive. For instance, Alashhab et al. (2024) proposed an ensemble online model by achieving 99.2% accuracy with minimal false alarms which is a common challenge in real world system. Building upon this, Jiyad et al. (2024) integrated ensemble learning with SHAP and LIME to provide interpretable results while maintaining a 97% detection accuracy. Tay et al. (2022) extended this work by applying Federated Learning with ensemble models across distributed datasets with privacy preservation, significantly improving generalization across different environments.

Despite their adaptive framework capabilities for responding to new attack pattern these approaches required computational resources and infrastructure, mainly in federated environments.

2.5 Importance of Feature selection in performance Optimization

Another area of focus in recent research has been feature selection, which significantly impacts the speed and accuracy of machine learning based DDoS detection. Ullah et al. (2023) and Farhat et al. (2023) both employed tree-based methods, Random Forest and Extra Trees respectively to identify and choose only high importance features. Their finding shows improvements in classification precision and accuracy. These embedded methods inherently consider feature interactions during model training, making them effective for complex network traffic datasets.

In contrast, Najar and Naik (2022) adopted Principal Component Analysis (PCA), a statistical technique that transforms original features into uncorrelated components, to reduce dimensionality. While PCA enhances computational efficiency, it may obscure feature interpretability, which is a critical limitation in security contexts.

Alduailij et al. (2022) further applied Random Forest Feature Importance (RFFI), aligning with the embedded approach but placing stronger emphasis on model transparency. For ensuring privacy without compromising the accuracy. Tay et al. (2022) integrated feature selection within a federated learning architecture, addressing privacy concerns in distributed environments. This comparison reveals a trade-off between accuracy, interpretability, and privacy, which suggests that the choice of feature selection method should align closely with the specific goals and constraints of the DDoS detection framework.

2.6 Dataset Dependence and Generalization Challenges

While reliance on standard datasets is useful for benchmarking, it may not work well when used on real, messy and unpredictable traffic. Most studies including Ullah et al. (2023), Farhat et al. (2023), and Najar and Naik (2022) used datasets like CICIDS2018, CICIDS2017, and UNSW-NB15 to evaluate model performance.

Bhardwaj et al. (2023), however, expanded on this by utilizing the BoT-IoT dataset, offering insights into DDoS behavior in IoT contexts. Alduailij et al. (2022) took this further by testing cross-dataset generalization, an essential step toward evaluating real-world applicability. These studies underline a critical challenge as models are trained on clean, well-labelled datasets may underperform on noisy, imbalanced real-world data.

2.7 Limitations and Research Gaps

Despite the significant progress, several gaps still exist particularly in handling real-time detection under evolving attack patterns. Many studies including Ullah et al. (2023), Al-Eryani et al. (2023), and Rajendran et al. (2023) identifies issues such as high computational demand, lack of diverse training data and difficulty in achieving real time performance at scale. Tay et al. (2022) and Bhardwaj et al. (2023) emphasized the need for lightweight, generalizable models suited for distributed environments and constrained devices.

To address these challenges, future research should focus on developing adaptive, resource-efficient models with strong generalization capabilities, supported by more realistic datasets and embedded interpretability. Furthermore, explainability remains an unresolved issue as most high performing model offer little insight into their decision-making process, making it more complicate to integrate into operational environment where accountability and auditability are essential.

2.8 Research Relevance and Importance of Literature Review

This literature review provides a comprehensive understanding of the current state of ML- based DDoS detection, identifying both achievement and challenges. This review is critical to this project as it informs the selection of supervised learning algorithms and the use of Random Forest Feature selection to improve both accuracy and computational efficiency. By analysing recent search, this project address key gaps such as real time adaptability, resource efficiency and need for interpretable models. Ultimately, this review ensures that the proposed system in this project is built on a solid foundation of proven methods while targeting existing weakness in current approaches

Chapter3: Requirements

To understand how machine learning can address the growing need for accurate and efficient DDoS attack detection in network security, the system requirements must first be clearly defined. This chapter outlines the functional and non-functional requirements of the proposed machine learning-based DDoS attack detection system. These requirements were identified through extensive research and prioritised using the MoSCoW method, a widely adopted technique for managing project scope.

According to ProductPlan (2024), the MoSCoW method classifies requirements into four categories:

Must have: If this requirement is not complete then the project is considered a failure.

Should Have: Its important but not critical but would provide excellent value to project if implemented.

Could Have: desirable requirements that are nice to have but not essential.

Won't have: Requirements that will not be included in the current timeframe but may be considered for the future.

3.1 Functional Requirements

Functional requirements define the specific operations and behaviours that the system must perform to detect DDoS attacks using machine learning techniques. The table below presents the prioritised functional requirements using the MoSCoW method.

ID	Requirement	MoSCoW	Description
FR1-	Data collection and preprocessing	Must have	The system must collect network traffic from publicly available datasets such as CICIDS2017, CICDDoS2019, etc. The dataset should be cleaned and pre-processed by handling missing values, normalization, and removing unnecessary fields
FR2	Feature Selection and Extraction	Must have	Must select and extract relevant feature for DDoS attack detection
FR3-	Model training time	Must have	Every model should display its training time, helping assess its suitability for real-world applications.
FR4	Performance Evaluation	Must Have	The system must evaluate the efficiency of the model using key metrics such as accuracy, precision, recall, and F1-score.
FR5	Model Comparison	Should Have	The system should compare multiple machine learning models to determine the most effective one for DDoS detection.
FR6	Hyperparameter Tuning	Should Have	The system should optimize model performance using hyperparameter tuning techniques such as GridSearchCV or RandomizedSearchCV.
FR7	visualization	Should Have	The system should provide graphical representations (e.g., confusion matrix, ROC curve, and feature

			importance) for better interpretation.
FR8	Anomaly Detection	Could have	The system could incorporate an anomaly detection model alongside Random Forest for hybrid detection.
FR9	Automated incident response	Won't have	The system will not automatically block malicious traffic it will only detect and alert
FR10	Real time detection capability	could have	System could analyse live network traffic data and classify it as normal or attack traffic.
FR11	Full intrusion detection system Features	Won't have	System will not act as full IDS but focus only on DDoS.
Fr12	Online Learning	Could have	The system could implement online learning to dynamically update the model in response to new attack patterns.

Table 1 : functional requirement

3.2 Non-functional Requirements

Non-functional requirements specify the quality attributes of the system, such as scalability, reliability, and accuracy. These requirements ensure the system performs efficiently and is suitable for practical deployment

Functional Requirement	Requirement	MoSCoW	Description
NFR-1	Scalability	Must have	The system must be able to handle large volumes of network traffic efficiently.
NFR-2	Reliability	Must have	The system must provide consistent and accurate ddos attack detection
NFR-3	Cloud Compatibility	Should Have	The system should be deployable on cloud environments like AWS, Azure, or Google Cloud
NFR-4	Implementation on Firewall	Could have	Can be implemented in the firewall for generating the alerts.
NFR-5	Accuracy	Must have	The system must achieve a minimum detection accuracy of 95% for DDoS attacks
NFR-6	Accuracy and Precision	Must have	Must maintain a high accuracy rate in detecting ddos attack with minimal false positive
NFR-7	Low Latency	Must have	Must process and classify traffic in real time with minimal time.
NFR-8	Physical Hardware Requirements	Won't have	Wont mandate specialized hardware for initial implementation
NFR-9	Maintainability	Could have	The system should be designed for easy maintenance and updates.

Table 2: non-functional requirements

Chapter4: Methodology

This section outlines the steps of the proposed methodology for DDoS attack detection. First, the CICIDS 2017 dataset is extracted and loaded. In the second step, preprocessing is performed to prepare the data for analysis. In the third step, machine learning algorithms are applied to classify DDoS attacks. Finally, the models are evaluated using various performance metrics to determine the most effective algorithm. Figure 2 illustrates the workflow of the proposed methodology for DDoS attack detection. Each phase of the methodology is detailed in the following subsections.

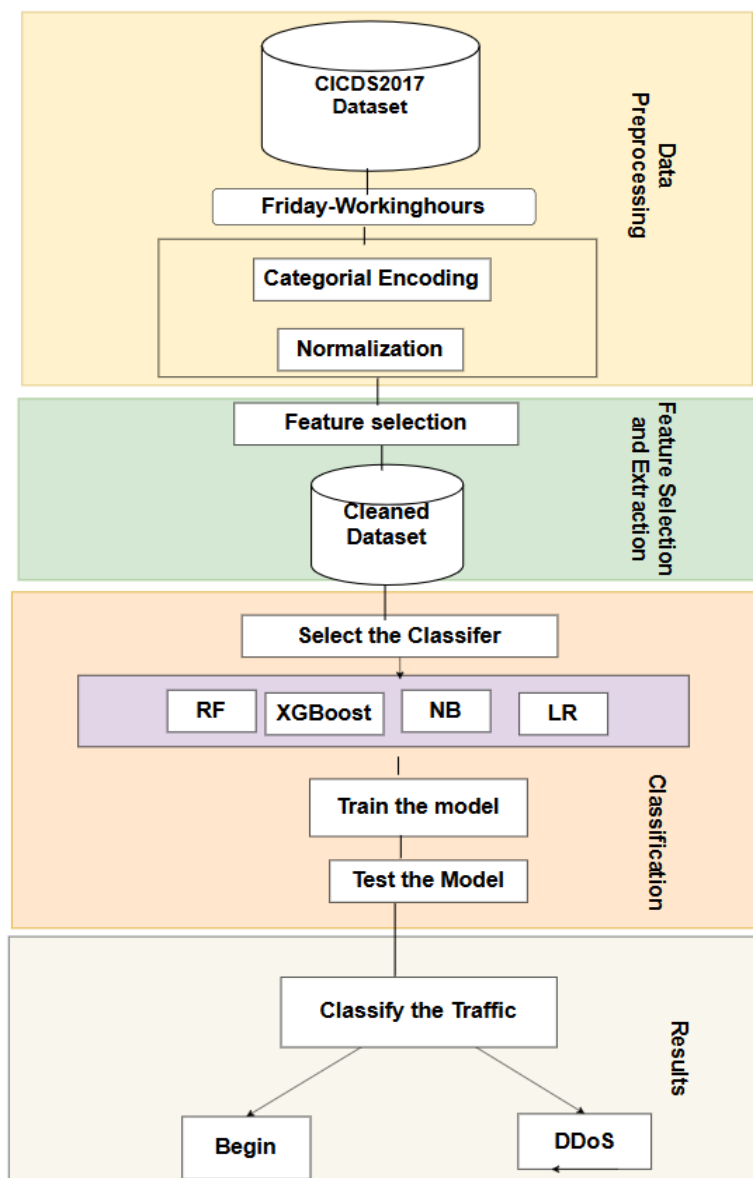


Figure 2: Architecture of the DDoS attack detection model.

Building a DDoS Detection Model

1.Dataset

To build Machine learning based DDoS attack detection system a reliable up to date label data set is required. There are many datasets that researchers used to evaluate the performance of their Intrusion detection system. Some of them are: KDDCUP99, NSL-KDD, CICIDS2017, CICDDoS2019 etc.

In this paper, CICIDS2017 dataset is used generated by the Canadian Institute of Cybersecurity as this dataset satisfies all the criteria of a reliable benchmark dataset (Sharafaldin et al., 2018). This CICIDS2017 dataset contains the data for the eight days available in a comma-separated values (CSV) format. It contains an abstract normal behaviour of twenty-five users based on the HTTP, HTTPS, FTP, SSH, and email protocols with several different attack traces, including Brute Force FTP, Brute Force SSH, DoS, Heartbleed, Web Attack, Infiltration, Botnet, and DDoS. To focus on DDoS attack detection FridayWorkingHoursAfternoonDDoS.pcap_ISCX.csv was chosen as it contains both normal and attack traffic. Additionally, it provides labelled flows and detailed network flow features extracted using CIC flow meter, which is suitable for supervised learning. It contains the seventy-eight feature and is labelled as benign or DDoS.

2.Data Preprocessing

In this phase, the dataset is first exported, then the features in the dataset are examined, where 78 features were found. Then data preprocessing was performed by eliminating rows containing infinity or Not a Number (NaN) values, as machine learning classifiers cannot process them in training or testing phases. To prepare the data for use in the Intrusion Detection System (IDS), two key preprocessing operations are applied.

2.1 Categorical Encoding:

As dataset contains only one non-numeric feature namely "Label" corresponding to "BENIGN" and "DDoS". Machine learning does not distinguish between different categories of attacks, the attributes "BENIGN" are converted to "0". Whereas the attributes corresponding to the "DDoS" are converted to "1". Figure 3 shows the distribution of traffic after categorical encoding.

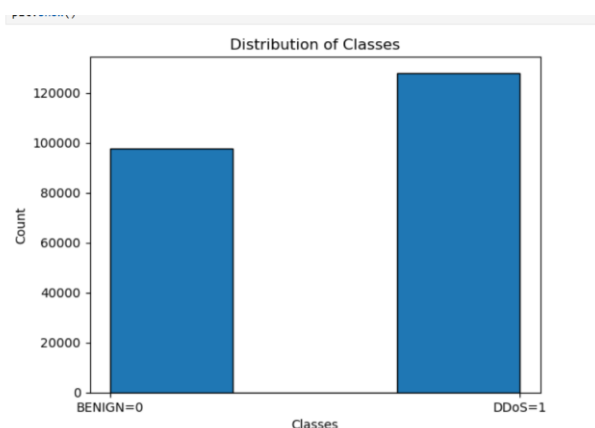


Figure 3: distribution of normal and DDoS traffic in Dataset.

2.2 Normalization:

We used the process of normalization known as the statistical Z-score method as it helps reduce classification errors significantly and allows the model to converge faster. The Statistical normalization (Z-score) formula is as follows:

$$X = (x - \mu) / \alpha \quad (1)$$

The value x of a feature A is transformed in X according to formula (1). μ is the mean and α is the standard deviation of a given attribute.

3. Feature Selection and Extraction

As the dataset is high dimensional, and the high-dimensional data take a lot of time for training. The goal of feature selection is to find the best feature subset of k features, which cause the least amount of generalization error (Alduailij *et al.*, 2022).

Here, Random Forest Feature Importance Method (RFFI) RF is used. It is an ensemble-learning algorithm that grows many decision trees, independently, and combines the output. Decision trees consist of internal and leaf nodes. The selected features are used to decide in the internal node, and it divides the dataset into two separate sets, with similar responses. The features in an internal node are selected by the Gini impurity criterion. The feature that has the highest decrease in impurity is selected for the internal node (Saeys, Abeel and Van de Peer, 2008).

Using the random forest feature importance method, feature extraction was done and drop all the other features that are not important as their presence will consume more overhead for loading and processing. Figure 4 shows the feature importance for DDoS detection for CICID 2017 dataset.

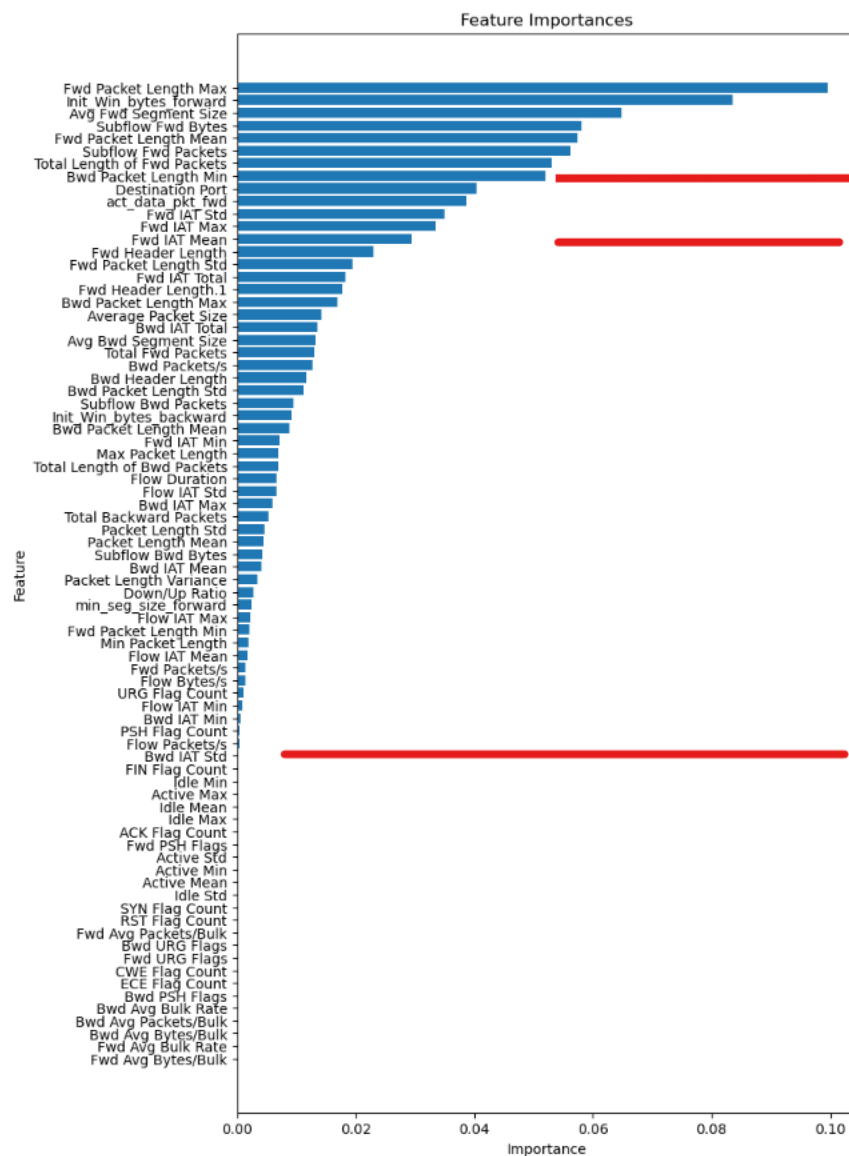


Figure 4 : Feature selection using random forest feature Importance.

4. DDoS Attack Classification

In this phase, the dataset was divided into 80% for training and 20% for testing. Four machine learning algorithms were applied: Random Forest, Logistic Regression, XGBoost, and Naive Bayes. These algorithms were chosen for their complementary strengths, balancing accuracy, computational efficiency, and adaptability to the dataset's structure.

Overview of Key Algorithms

4.1 Random Forest (RF)

This ensemble method uses decision trees to deliver accurate predictions by majority voting in classification tasks (Svetnik et al., 2003). RF builds multiple trees on randomized subsets of data

and features, reducing overfitting risks while improving robustness. It's highly efficient with large datasets and handles many input variables without requiring feature reduction (Chen et al., 2020).

4.2 XGBoost

A sophisticated gradient boosting algorithm recognized for its efficiency and scalability. Developed by Chen and Guestrin (2016), XGBoost uses advanced regularization techniques to optimize training speed and accuracy, making it a favorite for competitive machine learning tasks.

4.3 Naive Bayes (NB)

Based on Bayes' Theorem, this probabilistic classifier assumes conditional independence between features. While the "naive" assumption might not always hold true, NB is fast, interpretable, and effective for various classification tasks.

4.4 Logistic Regression (LR)

A robust method for binary classification that optimizes weights using the lbfgs algorithm. Logistic Regression processes input through a sigmoid activation function, delivering precise predictions (Saeys, Abeel and Van, 2008).

Rationale for Model Selection

These algorithms were chosen for their diverse strengths in tackling classification problems. Random Forest and XGBoost provide high accuracy and resilience against overfitting through ensemble techniques. Logistic Regression and Naive Bayes offer simplicity and speed, serving as strong baseline models. Meanwhile, MLP contributes its ability to model complex, non-linear data relationships. Compared to resource intensive methods like deep learning or SVMs, the selected models strike a balance between performance and efficiency, essential for real-time DDoS detection.

5. Performance Metrics for Evaluation of models

The effectiveness of an Intrusion Detection System (IDS) is primarily measured by its ability to make accurate predictions. A common approach to evaluate this is by comparing the IDS's predictions against the actual outcomes of events. This comparison results in four outcomes, which are organized in a confusion matrix, as shown in Table 3.

5.1 Confusion Matrix

According to GeeksforGeeks (2020), the confusion matrix is a tabular representation that categorizes predictions into the following four outcomes:

- **True Positives (TP):** Instances where the IDS correctly predicts an event as Malicious/DDoS, and it is indeed Malicious/DDoS.
- **True Negatives (TN):** Instances where the IDS correctly predicts an event as Benign, and it is actually Benign.

- **False Positives (FP):** Instances where the IDS incorrectly predicts an event as Malicious/DDoS, but it is actually Benign.
- **False Negatives (FN):** Instances where the IDS incorrectly predicts an event as Benign, but it is actually Malicious/DDoS.

To optimize the performance of an IDS, it is essential to maximize TP and TN rates, while minimizing FP and FN rates.

Actual class	Predicted class	
	DDoS	Benign
DDoS	True positive (TP)	False Negative (FN)
Benign	False Positive (FP)	True Negative (TN)

Table 3: confusion matrix

5.2 Performance Metrics

According to GeeksforGeeks (2020), the confusion matrix is a tabular representation that categorizes predictions into the following four outcomes: following evaluation metrics are commonly used to assess the performance of classification models, including IDS. These metrics are independent of the training and testing dataset size and provide deeper insight into the model's ability to correctly detect threats

5.2.1 Accuracy

Accuracy is the proportion of total correct predictions (both TP and TN) to the total number of predictions. It reflects how often the model is correct overall.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

5.2.2 Precision

Precision measures the proportion of correctly predicted DDoS instances (TP) out of all instances predicted as DDoS (TP + FP). It helps assess how many detected threats are actually malicious.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

A high precision value indicates a low false positive rate, which is important in minimizing false alarms.

5.2.3 Recall

Recall evaluates how effectively the model identifies actual DDoS attacks. It is the ratio of true positive predictions to all actual DDoS instances (TP + FN).

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

High recall means the IDS is good at detecting most of the malicious activities.

5.2.4 F1-score

F1-Score is the harmonic mean of Precision and Recall. It provides a balanced metric when there is an uneven class distribution or when both FP and FN are important to consider.

$$\text{F1-Score} = 2 \times \text{Recall} \times \text{Precision} / (\text{Recall} + \text{Precision}) \quad (4)$$

F1-Score is especially useful when a balance between false positives and false negatives is needed.

5.2.5 False Alarm Rate (FAR)

False Alarm Rate (FAR) is the proportion of benign instances incorrectly classified as malicious. It reflects the likelihood of false positives.

$$\text{False Alarm Rate} = \text{FP} / (\text{FP} + \text{TN}) \quad (5)$$

A lower FAR indicates better reliability of the IDS in avoiding unnecessary alerts.

Chapter 5: Result and Discussion

The main goal of this project was to evaluate the performance of different learning model such as Logistic Regression (LR), Random Forest (RF), Naive Bayes (NB), and XGBoost in detecting DDoS attacks. DDoS attack detection is a binary class problem, with traffic begin either such as benign or DDoS attack. To improve the classification, RFFI was utilized to select the most relevant features for detecting DDoS attack, and the evaluation was carried out on four different feature sets: all features, 53 features, 13 features and 8 features and the result can be seen in table 5. The detail of the experiment setup for this project is present in table 4.

Components	Description
Hardware	Intel core i9 13 th gen
Operating system	Windows 11
Memory	16 GB RAM
Libraries	Pandas, SK learn, Matplotlib
Core programming language	Python
IDE	(Anaconda Jupyter Notebook)

Table 4: Details of the experiment setup used.

Model Performance Overview

The results from Table 5 show that XGBoost consistently outperformed other models, achieving 100% accuracy, precision, and recall, along with a 0.01% false alarm rate when using all features and the 53 features identified by Random Forest feature importance. This indicates that XGBoost is highly effective in correctly identifying DDoS attacks with almost no errors. Meanwhile, Random Forest also performed exceptionally well, achieving 99.99% accuracy and 100% precision with both all features and the 53 features. However, its false alarm rate was slightly higher compared to XGBoost.

Whilst Logistic Regression AND Navie Bayes show a significant decline in performance when fewer features were used. For example, with 8 features, LR false alarm area increased to 32.20% and NB's false alarm rate reached 24.54%. This suggests show that LR and NB required more comprehensive set of features to maintain their performance, whereas XG boost, and RF handle reduced feature sets effectively.

Impact of feature selection

In this experiment, Feature selection played a critical role in the performance of the model. While using all features generally all models gave a best result with, with high accuracy, precision, recall, and low false alarm rates. But when the feature set was reduced to 53 features, the performance of XGBoost and RF remained almost the same as with all features, indicating that these models can effectively handle a reduced feature set without significant loss in performance.

However, with 13 features, XGBoost and RF maintained impressive performance, but LR and NB showed a noticeable drop in accuracy and precision, along with a significant increase in false alarm rates. With 8 features, LR and NB struggled the most, with LR showing a false alarm

rate of 32.20% and NB showing 24.54%. This suggests that these models are more sensitive to the number of features and require a more comprehensive feature set for optimal performance.

Training Time for models

Training time plays the crucial role of in the real world while detecting the DDoS attack as it is important while developing and maintaining and accurate and update model. Although the training time varies every time the system is run but it shows similar result. In terms of training time, NB was the fastest taking less minutes compared to other model across all feature sets making it most computationally efficient model, which can be good fit in scenario where quick training is required. Logistic Regression (LR) and XGBoost also had relatively low training times, with LR taking 0.04 minutes and XGBoost taking 0.07 minutes with all features. These models have a good balance between performance and computational efficiency. On other hand, RF had the longest training time are 0.78 mins with all features. Whiles this is slower compared to the other models, it still valuable given its high performance where it can be used where accuracy and precision are important. Overall, the training times for all models were relatively low, making them suitable for real world applications where efficiency is important.

Model Name	Accuracy	F1-score	Precision	Recall	False Alarm rate	Training time
With all features						
RF	99.99%	99.99	100%	99.99 %	0.00%	0.78 min
LR	99.86%	99.86	99.89	99.88	0.15%	0.08 min
NB	98.81	98.97	99.08	98.86	2.59	0.04 min
XGBoost	100	100	100	100	0.01	0.07 min
With 53 Feature Selection						
RF	99.99	99.99	100	99.99	0.00	0.25min
LR	99.76	99.79	99.71	99.87	0.38	0.02 min
NB	98.76	97.92	99.98	98.88	2.73	0.00 min
XGBoost	100	100	100	100	0.01	0.03 min
With 13 feature selection						
RF	99.98	99.98	100	99.97	0.01	0.10 min
LR	96.58	97.08	94.52	94.52	7.67	0.01 min
NB	97.91	98.20	96.49	99.97	4.81	0.00 min
XGBoost	99.98	99.98	99.99	99.97	0.01	0.04 min
With 8 feature selection						
RF	99.94	99.94	99.96	99.92	0.05	0.04min
LR	86.13	89.15	80.44	99.97	32.20	0.05 min
NB	89.42	91.51	84.37	98.97	24.54	0.00 min
XGBoost	99.93	99.94	99.96	99.92	0.05	0.01min

Table 5: Model performance of Different algorithms using Evaluation metrics

Confusion metrics

Table 6 shows the confusion matrix of different machine learning models for cids 2017 dataset. The results show true negative rate, false positive rate, false negative rate, and true positive rate, with different feature sets. The purpose of confusion matrix is to show the miss classification rate of all methods. Low miss classification reflects better accuracy. All these methods have high accuracy and other metrics values. But for large dataset, measuring accuracy only is not a valid measurement, since miss classification of some data points does not affect the accuracy (mona etl 2022). The confusion matrix provided a detailed breakdown of how well each model performed in classifying the traffic. XGBoost achieved 19418 True Negatives (TN), 1 False Positive (FP), 0 False Negatives (FN), and 25724 True Positives (TP) with all features, indicating near-perfect classification. This means that XGBoost correctly identified almost all DDoS attacks and benign traffic, with almost no misclassifications. Random Forest (RF) also performed very well, with 19419 TN, 0 FP, 3 FN, and 25721 TP using all features. This demonstrates that RF is highly effective in detecting DDoS attacks with minimal errors.

However, Logistic Regression (LR) and Naive Bayes (NB) showed mixed results. While LR performed well with all features and 53 features, its performance degraded significantly with fewer features. For example, with 8 features, LR had 13166 TN, 6253 FP, 9 FN, and 25715 TP, indicating a high rate of false positives. Similarly, NB had the highest false positives across all feature sets. With 8 features, it achieved 14654 TN, 4765 FP, 9 FN, and 25715 TP, suggesting that NB is less reliable for distinguishing between benign and DDoS traffic, especially when fewer features are used. This highlights the importance of feature selection for models like LR and NB, which struggle to maintain performance with reduced feature sets.

Method	True Negative	False Positive	False Negative	True Positive
With All features				
XGBOOST	19418	1	0	25724
NB	18917	502	35	25689
LR	193390	29	32	25692
RF	19419	0	3	25721
With 53 features				
XGBOOST	19418	1	0	25724
NB	25693	530	31	25693
LR	19345	74	34	25690
RF	19419	0	3	25721
With 13 features				
XGBOOST	19417	2	7	25717
NB	18484	935	9	25715
LR	17930	1489	53	25671
RF	19418	1	8	25715
With 8 Features				
XGBOOST	19409	10	21	25703
NB	14654	4765	9	25715
LR	13166	6253	9	25715
RF	19410	9	20	25704

Table 6: confusion matrix reference to figure 10,11,12,13

Roc Curve under Accuracy

The ROC curve under the accuracy performance shows near perfect classification across several models. However, the persistent AUC= 1.0 score obtained by both Random Forest and XGBoost raised concerns about overfitting or potential data leakage regardless of feature drop including 8, 13, 53 and full feature set. This suggest that the model may have accidentally learned from label features or leaked information. In contrast, Logistic Regression shows a slightly decline in performance with its AUC dropping from 0.9998 with full features to 0.9790 when limited to 8 features, showing dependency to feature reduction. Meanwhile, Naïve Bayes maintained a stable AUC of approximately 0.985 regardless of the feature set, suggesting that many of the original features may be redundant and do not significantly influence the model's predictive ability. Overall, while the result demonstrates high performance, the possibility of overfitting is high.

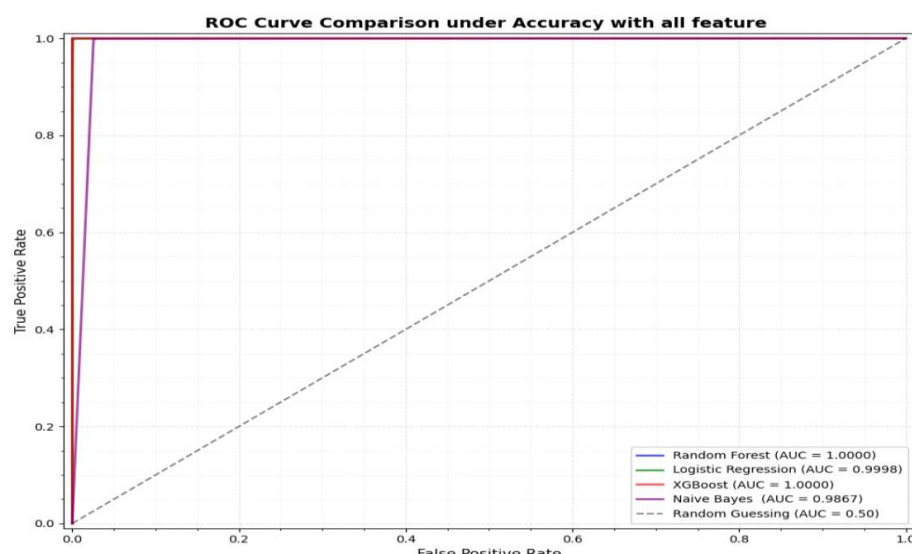


Figure 5: Roc curve with all feature set

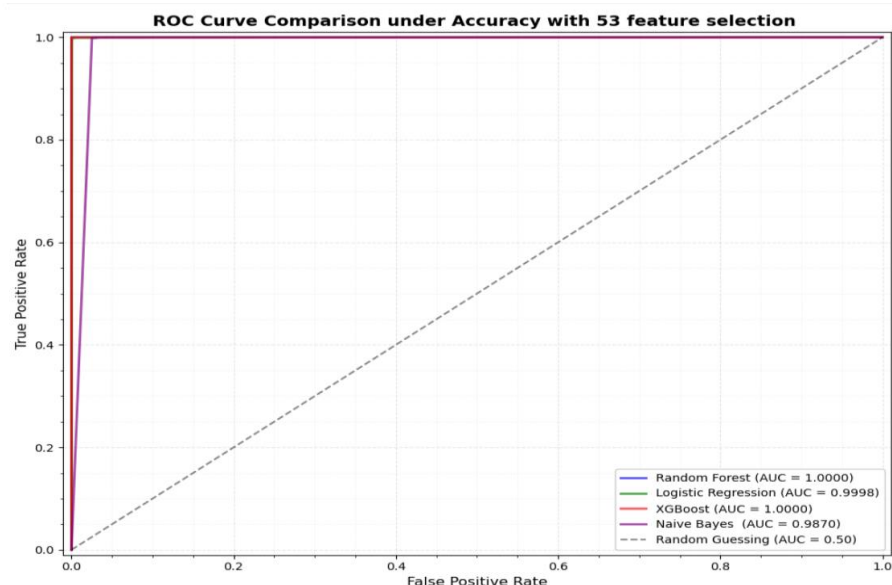


Figure 6: ROC curve with 53 feature set

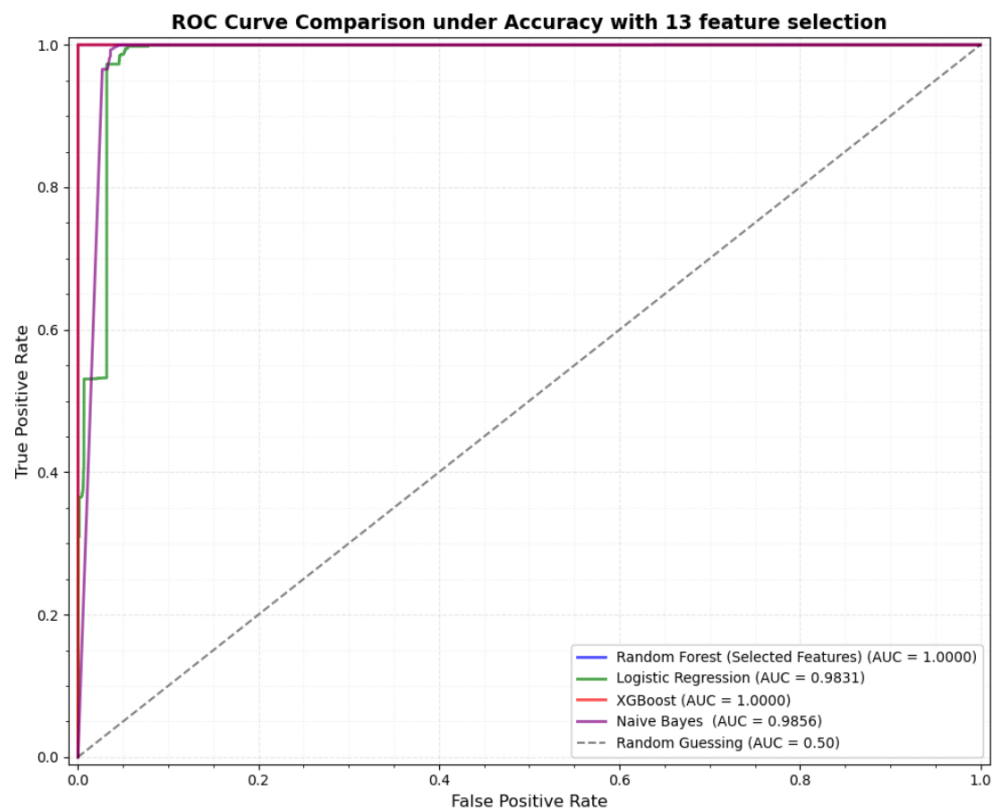


Figure 7: ROC curve with 13 feature set

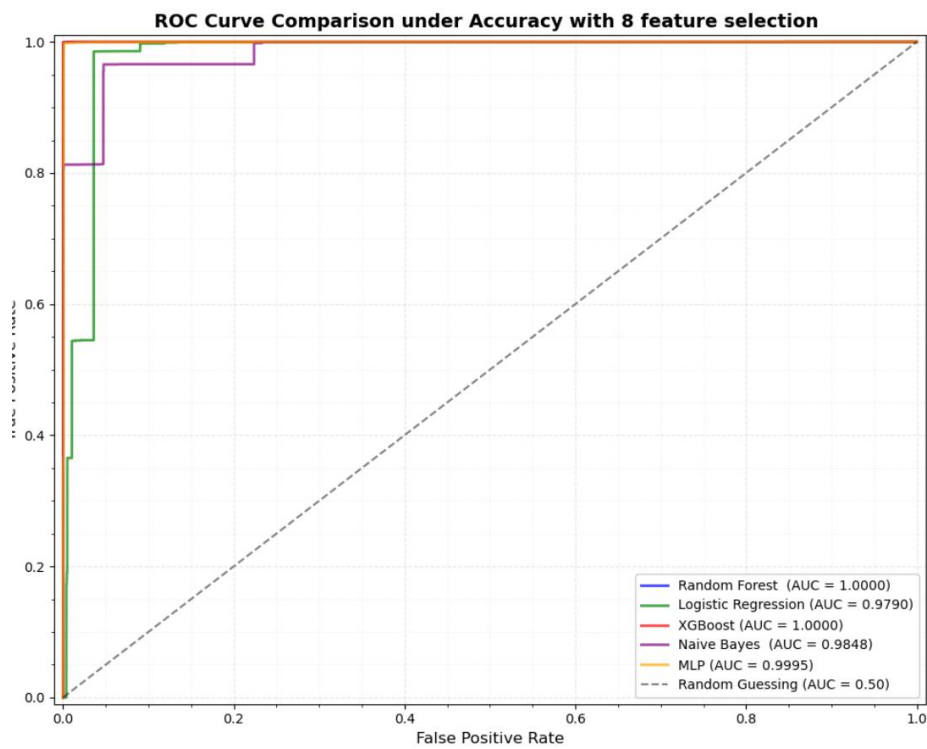


Figure 8: ROC curve with 8 feature set

Use of Stratified K-Fold Cross-Validation to Assess Overfitting

To assess the model's robustness and generalizability, stratified 5-fold cross validation was used using all feature set. In this project, stratified cross-validation was applied to four models (Random Forest, XGBoost, Logistic Regression, and Naive Bayes). Each model was trained and validated across different stratified subsets of the training data, ensuring that the class distribution was preserved in each fold. Performance metrics such as accuracy, precision, recall, and F1-score were averaged across the five folds to obtain a more comprehensive evaluation.

For instance, Random Forest and XGBoost both achieved 99.99% across all metrics as can be seen in table 7, indicating highly stable performance. Logistic Regression and Naive Bayes also demonstrated strong results, achieving mean accuracies of 99.87% and 98.78%, respectively.

This approach provides a more accurate idea of how well the model works by testing it on different parts of the data instead of just one split ('Stratified K Fold Cross Validation', 2020). This proves that model is not just doing well on a luck or easy dataset. It confirms that the models, especially ensemble ones, are not just performing well on one specific partition but consistently across various subsets. Therefore, Stratified K-Fold Cross-Validation enhances the credibility of the performance results and strengthens confidence in the models' applicability for real-world DDoS detection tasks.

Model	Accuracy (%)	Precision (Macro) (%)	Recall (Macro) (%)	F1 Score (Macro) (%)
Random Forest	99.99	99.99	99.99	99.99
XGBoost	100	99.99	100	99.99
Logistic Regression	99.87	99.87	99.86	99.86
Naive Bayes	98.78	98.93	98.6	98.76

Table 7 Stratified K-Fold Cross validation Result(5-fold) with all features

Sanity Check Using Shuffled Labels

To validate that the high performance of the models was not because of overfitting, data leakage, or spurious correlations, a sanity check is performed by randomly shuffling the training labels and rerunning cross validation (Scikit-learn, 2019; 'sklearn.utils.shuffle', n.d.).

Each classifier was evaluated using 5-fold cross validation on the training set with randomly permuted labels. The result show drastically dropped of the model performance with mean cross validation accuracies close to the random guessing, Random Forest (52.18%), XGBoost (55.69%), Logistic Regression (56.65%), and Naive Bayes (56.18%) as seen in figure 14,15,16,17. These results are close to random guessing, as the models could no longer identify meaningful patterns in the data.

This sharp decline in accuracy validates that the original high performance of the models was due to genuine learning rather than overfitting or data leakage. It reinforces the integrity of the experimental setup and demonstrates that the features selected for model training are indeed predictive of DDoS attack activity. Consequently, this sanity check increases confidence in the robustness and reliability of the classifiers developed in this study.

Key Finding

The key finding of this project was XGBoost and Random Forest were the best models for DDoS detection, with XGBoost slightly outperforming RF in terms of false alarm rate and training time. Both models achieved near-perfect accuracy and precision, making them highly dependable for detecting DDoS attacks. Second, feature selection plays a crucial role in model performance. While XGBoost and RF can handle reduced feature sets effectively, Logistic Regression (LR) and Naive Bayes (NB) require more features to maintain their performance. Reducing features too aggressively (e.g., to 8 features) significantly degrades the performance of LR and NB, leading to higher false alarm rates.

Third, false alarm rates are a critical metric in DDoS detection. High false positives can lead to unnecessary alerts and actions, which can be costly in real-world scenarios. XGBoost and RF excel in minimizing false alarms, while LR and NB struggle, especially with fewer features. Finally, training time is low for all models, with NB being the fastest and RF being the slowest. However, the difference in training time is minor compared to the performance gains achieved by XGBoost and RF.

Furthermore, to ensure the generalizability and robustness of the models, a Stratified 5-Fold Cross-Validation approach was employed. This method provided a reliable estimate of model performance across different data subsets, confirming that the high scores were not the result of a favorable train-test split, but rather reflected consistent performance across multiple partitions.

Finally, a sanity check was conducted by randomly shuffling the training labels and repeating the cross-validation process. As expected, model performance dropped significantly to near-random levels (e.g., approximately 52% accuracy for Random Forest), which confirmed that the original high performance was due to meaningful learning and not overfitting or data leakage. This test strengthened the credibility of the results and validated the effectiveness of the chosen features and models for real-world DDoS detection tasks.

Chapter 6: Project Evaluation

This aim of the project was to evaluate the performance of four machine learning models: Logistic Regression (LR), Random Forest (RF), Naive Bayes (NB), and XGBoost for detecting DDoS (Distributed Denial of Service) attacks. The results showed that XGBoost and RF are the most effective models, with XGBoost slightly outperforming RF in terms of false alarm rate and training time. Both models achieved near-perfect accuracy and precision, making them highly reliable for real-world DDoS detection.

The project also explored the impact of feature selection by evaluating the models on different feature sets (all features, 53 features, 13 features, and 8 features). The results showed that XGBoost and RF can handle reduced feature sets effectively, while LR and NB require more features to maintain performance. This highlights the importance of careful feature selection, especially for models like LR and NB, which struggle with fewer features.

To ensure robustness and prevent overfitting, the Stratified 5-Fold Cross-Validation was chosen, which maintains the class distribution across all folds. This approach provided more accurate and stable estimates of model performance across different subsets of the training data, strengthening the reliability of the reported metrics. Models such as XGBoost and RF consistently performed well across all folds, reinforcing their generalizability.

Additionally, a Sanity Check was conducted by randomly shuffling the training labels and re-evaluating model performance using cross-validation. The resulting accuracy scores dropped significantly for all models hovering around 50–56%, as expected for random guessing on a balanced dataset. This confirmed that the high performance of the models in the original experiments was due to learning meaningful patterns in the data, rather than overfitting or data leakage.

The project has several strengths that contributed to its success. First, it conducted a comprehensive evaluation of multiple machine learning models, including Logistic Regression (LR), Random Forest (RF), Naive Bayes (NB), and XGBoost, using key metrics such as accuracy, precision, recall, and false alarm rate. This thorough analysis provided a clear understanding of each model's performance in detecting DDoS attacks. Second, the project included a detailed feature selection analysis, examining how reducing the number of features impacted model performance. This highlighted the importance of feature selection, especially for models like LR and NB, which struggled with fewer features. Third, the project offered practical recommendations, guiding the selection of the best model based on accuracy, false alarm rates, and training time, making the findings highly applicable to real-world scenarios.

However, the project also has some limitations. The models were tested on a single dataset, which may limit the generalizability of the results. Additionally, advanced feature engineering techniques were not extensively explored, which could have further improved model performance. The project also focused solely on binary classification (DDoS vs. Benign) and did not address different types of DDoS attacks, such as volumetric or application-layer attacks.

To address these limitations, several future improvements can be made. Evaluating the models on multiple datasets would ensure the results are more generalizable and robust. Exploring advanced

feature engineering techniques, such as domain-specific feature extraction or automated feature selection, could further enhance model performance. Implementing the models in real-time DDoS detection systems would provide insights into their effectiveness in live environments. Finally, extending the project to include multi-class classification would allow for the detection of diverse types of DDoS attacks, improving the overall effectiveness of the detection system. These improvements would build on the project's solid foundation and further advance the field of DDoS detection using machine learning.

Chapter 7: Further Work and Conclusions

While the results of this project demonstrate the effectiveness of machine learning models in detecting DDoS attacks, there are several areas to improve further. One important extension would be to move beyond binary classification and implement multi-class classification models that can differentiate between several types of DDoS attacks, such as volumetric, protocol-based, and application-layer attacks. This would provide more granular insights and improve the response strategies in real-world systems. Additionally, integrating the models into real-time network environments would offer practical insights into their performance under live conditions, including latency, throughput, and scalability. Another valuable direction involves testing the models on diverse datasets to assess their generalizability across different network traffic scenarios and attack patterns. Advanced feature engineering techniques, such as domain-specific feature extraction or automated methods like principal component analysis (PCA), could also be explored to further enhance performance and reduce computational overhead. Moreover, future work may investigate the robustness of these models against adversarial techniques, such as traffic obfuscation or evasion attacks, which are becoming increasingly prevalent in modern cyber threats. Lastly, applying model interpretability tools, such as SHAP values, and conducting hyperparameter optimization through techniques like GridSearchCV, would improve both the transparency and performance of the models, enabling better trust and adoption in operational settings.

In conclusion, this project has shown that XGBoost and Random Forest are highly effective models for detecting DDoS attacks, with XGBoost slightly outperforming Random Forest in terms of false alarm rate and training efficiency. Both models maintained high accuracy and precision, even when evaluated using reduced feature sets, highlighting their robustness and reliability. In contrast, Logistic Regression and Naive Bayes exhibited significant performance degradation with fewer features, underscoring the importance of careful feature selection. The use of stratified cross-validation ensured that the evaluation was statistically reliable across balanced class distributions, while the implementation of a sanity check—achieved by shuffling class labels and re-evaluating model performance—validated that the models were genuinely learning meaningful patterns rather than overfitting to noise. Despite its limitations, such as the use of a single dataset and the binary classification focus, the project provides a solid foundation for future research and practical applications. By building upon these results, future work can advance the development of more intelligent, adaptable, and real-time DDoS detection systems using machine learning.

Glossary

Intrusion Detection System (IDS): A software or hardware system that monitors network or system activities for malicious actions or policy violations.

Machine Learning (ML): A subset of AI where algorithms learn from data patterns to make predictions or decisions without being explicitly programmed.

XGBoost: A high-performance machine learning algorithm based on gradient boosting, known for its accuracy and efficiency in classification tasks.

UNSW-NB15 Dataset: A modern dataset developed by the University of New South Wales for evaluating network intrusion detection systems.

NSL-KDD Dataset: A refined version of the KDD'99 dataset used to evaluate the performance of IDS models.

Naive Bayes (NB): A simple probabilistic classifier based on Bayes' Theorem with an assumption of independence among predictors.

Support Vector Machine (SVM): A supervised machine learning model used for classification and regression tasks.

Security Operations Center (SOC): A centralized unit that monitors, detects, and responds to cybersecurity threats in an organization.

Principal Component Analysis (PCA): A dimensionality reduction technique used to simplify large datasets while preserving important information.

False Positive Rate (FPR): The proportion of normal traffic incorrectly identified as an attack by a detection system.

Botnet: A network of compromised computers controlled remotely by attackers to perform malicious activities, including DDoS attacks.

Artificial Intelligence (AI): A branch of computer science that enables machines to simulate human intelligence, including learning, reasoning, and problem-solving.

Stratified Cross-Validation: A type of cross-validation technique used in machine learning where the dataset is split into k folds in such a way that each fold has approximately the same percentage of samples of each target class as the complete dataset. This ensures that all classes are properly represented in both training and validation sets, especially important in imbalanced classification problems like DDoS detection.

Table of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
ANN	Artificial Neural Network
BoT	Botnet
CPU	Central Processing Unit
CSV	Comma-Separated Values
CyBOK	Cyber Security Body of Knowledge
DDoS	Distributed Denial of Service
DNS	Domain Name System
DoS	Denial of Service
FPR	False Positive Rate
IDS	Intrusion Detection System
IoT	Internet of Things
KNN	K-Nearest Neighbors
ML	Machine Learning
NB	Naive Bayes
NSL-KDD	Network Security Laboratory – Knowledge Discovery in Databases
NSO	NSO Group (Israeli cyber-intelligence firm)
PCA	Principal Component Analysis
RFC	Request for Comments
ROC	Receiver Operating Characteristic
RNN	Recurrent Neural Network
SOC	Security Operations Center
SVM	Support Vector Machine
TP	True Positive

TPR	True Positive Rate
UDP	User Datagram Protocol
UNSW-NB15	University of New South Wales - Network-Based 2015 Dataset
VPN	Virtual Private Network
XGBoost	Extreme Gradient Boosting
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
FTP	File Transfer Protocol
SSH	Secure Shell
DDoS	Distributed Denial of Service
FTP BF	Brute Force File Transfer Protocol
Web Attack	Web-Based Attack
SSH BF	Brute Force Secure Shell
Heartbleed	Heartbleed Vulnerability Exploit
Botnet	Robot Network
Infiltration	Unauthorized Network Intrusion

Appendix A: First Appendix

Meeting Logs

Date	Meeting Notes	Actions Taken / To Take
18th Nov 2024	- Supervisor reviewed my proposal and advised narrowing the scope, as it was too broad for the project timeline. - Discussion about focusing on the introduction, scope, aims, and objectives. - Advised to start working on the literature review.	- Drafted the introduction, aims, and objectives. - Changed objectives to focus on machine learning-based detection only. - Conducted literature review on 10+ key resources related to DDoS attacks and machine learning. - Used Research Rabbit to find relevant sources. - Decided to compare different machine learning models to identify the best-performing algorithm.
2nd Dec 2024	- Aims and objectives should follow the SMART (Specific, Measurable, Achievable, Relevant, Time-bound) technique. - Use evaluation metrics such as F1-score, recall, and accuracy. - Literature review needs to be completed by the next meeting.	- Focused on supervised learning algorithms (Random Forest, Logistic Regression, MLP). - Refined the literature review based on the project focus.
27th Jan 2025	- Identified functional and non-functional requirements. - Literature review should strictly focus on supervised learning (no need to include unsupervised methods).	- Updated literature review to align with the project scope. - Finalized functional and non-functional requirements.
10th Feb 2025	- Supervisor suggested sending the code for review.	- Began cleaning and refining the code for submission. - Focused on preparing a reviewable version.
24th Feb 2025	- Recommended using Pickle for faster file loading. - Compare model execution times to assess real-world applicability. - Analyze feature importance. - Discussed findings from confusion matrix analysis. - Discussed real-world applications and future vision of the project. - Emphasis on completing theoretical	- Implemented Pickle for model loading. - Compared model training and prediction times. - Analyzed feature importance. - Used Excel for detailed time management.

	sections quickly for review. - Suggested using Excel for time management.	
10th March 2025	- Discussion on feature selection techniques.	- Applied feature selection methods to improve model performance.
24th March 2025	- Reviewed literature review progress. - Supervisor mentioned Studiosity for academic writing support. - Encouraged to book a 1-to-1 session with the library team for citation guidance and improving critical writing.	- Booked and attended a 1-to-1 library session with Sahara. - Improved academic writing and citation techniques based on feedback.
9th April 2025	- Discussed the XGBoost model achieving 100% accuracy. - Emphasis on critical analysis of results. - Ensure data is clean: check for duplicate columns and constant features. - Perform K-Fold Cross-Validation. - Use smaller subsets if necessary. - Check for potential data leakage.	- Conducted critical evaluation of XGBoost results. - Cleaned dataset (removed duplicate columns, dropped constant features). - Implemented K-Fold Cross-Validation. - Verified no data leakage present. - Sanity check by shuffling the label feature and testing

Time management and Project planning

S.N.	Task	Start Date	Finish Date	Soft Deadline (Target)	Planning Notes
1	Topic Identification	16/10/2024	23/10/2024	20/10/2024	Brainstorm initial ideas; assess feasibility and relevance to CyBOK.
2	Topic Refinement	24/10/2024	30/10/2024	28/10/2024	Finalize focus on ML for DDoS detection using supervised learning.
3	Define Aims and Objectives	31/10/2024	4/11/2024	2/11/2024	Draft project aim, specific objectives, and project scope.
4	Write Introduction	5/11/2024	12/11/2024	10/11/2024	Outline motivation, context, and background information.
5	Literature Review	13/11/2024	30/3/2025	27/12/2024	Use ResearchRabbit; review recent ML-based DDoS detection studies.
6	Requirements Specification	1/12/2024	7/12/2024	5/12/2024	Define both functional and non-functional system requirements.
7	Design ML Workflow	8/12/2024	15/12/2024	13/12/2024	Plan architecture, tools, and pipeline; draw system workflow.
8	Data Collection	16/12/2024	3/1/2025	30/12/2024	Acquire and validate datasets (e.g., CICID2017).
9	Data Preprocessing	4/1/2025	11/1/2025	9/1/2025	Clean, encode, normalize, and prepare data for analysis.
10	Feature Selection	12/1/2025	18/01/2025	16/01/2025	Use correlation, Chi-square test, and Random Forest importance.
11	Model Selection	19/01/2025	24/01/2025	22/01/2025	Choose classification algorithms for experimentation.
12	Model Training: Random Forest	25/01/2025	29/01/2025	27/01/2025	Train and evaluate baseline model.
13	Model Training: Logistic Regression	30/01/2025	2/2/2025	1/2/2025	Train and evaluate.
14	Model Training: Naïve Bayes	3/2/2025	6/2/2025	5/2/2025	Train and evaluate.
15	Model Training: XGBoost	7/2/2025	11/2/2025	9/2/2025	Train and evaluate.
16	Model Comparison & Analysis	19/02/2025	25/02/2025	22/02/2025	Compare performance: accuracy, precision, recall, ROC-AUC.
17	Results Discussion & Evaluation	26/02/2025	5/3/2025	3/3/2025	Analyze findings, justify performance, and evaluate methods.
18	Conclusion & Future Work	6/3/2025	10/3/2025	8/3/2025	Summarize key insights and suggest future improvements.
19	Reference Management	11/3/2025	15/03/2025	13/03/2025	Organize sources using Zotero or Mendeley; check citation styles.
20	Document Refinement	16/03/2025	10/4/2025	6/4/2025	Proofreading, structure polishing, figure updates, citation checks.
21	Final Review & Corrections	11/4/2025	20/04/2025	17/04/2025	Gather supervisor feedback; make final changes.
22	Submission	—	30/04/2025	25/04/2025	Final upload to university submission portal.

Figure 9: Project Management

December Progress Review

Project Details	
Student Name:	SANISH SINGH MAHARJAN
Student Number:	22085549
Student Email Address:	Sanishs2.singhmaharjan@live.uwe.ac.uk
Date	December Progress Review
Supervisor Name	Jonathan White
Project Title:	Machine learning for DDoS attack Detection

Project Overview
- A brief description of the project aims and objectives (refined from the proposal) The aim of this project is to design, develop and evaluate a supervised machine learning model for the accurate and efficient detection of DDoS attacks using the feature selection in a network traffics. Using the label dataset such as CICIDS 2017, this model will focus on classify network traffic as normal traffic and DDoS attack, improving the accuracy whilst checking speed of the intrusion detection system (IDS) for real time use. Objectives - To conduct study on the working mechanism of DDoS attack and challenges for detection. - Exploring the current Machine learning detection models and choosing the best model for this project. - Using the CICIDS 2017 Dataset for Training and validating the selected machine learning, ensuring the proper application of preprocessing and feature selection technique - Using the random forest feature importance as the feature selection and using selected feature for training the other models. - Evaluating and comparison the popular models such as random forest, logistic regression etc. using the (accuracy, precision, recall, F1-score) - Use Receiver Operating Characteristics (ROC) CURVES AND Area Under Curve (AUC) metric for comparing the model performance.

Progress To Date
- Summary of work complete so far: Dissertation - Literature review: scope, quality and initial findings. Development of the theoretical framework or research questions Technical – Literature review, summary of requirements analysis, initial designs - Key milestones achieved - Challenges Encountered and how they were addressed
Ethical Approval Status - working draft of the literature review, focusing on existing ML approaches to DDoS detection, common datasets, and performance evaluation metrics. - Downloaded and explored the CICIDS 2017 dataset, noting its features, class labels, and imbalances. - Started investigating feature selection methods and ML models for later implementation. Key Milestones Achieved - Finalized project scope and narrowed objectives. - Drafted a structured literature review document. - Set up the working environment for coding and data preprocessing. - Identified Random Forest and Logistic Regression as primary models for evaluation Challenges Encountered - Early versions of the literature review lacked proper citations and structure. Solution: Supervisor feedback helped restructure the review using a thematic, Statement-Evidence-Evaluation (SSE) format with proper referencing. - The project scope was initially too broad. Solution: Refined the project focus based on feedback to concentrate strictly on supervised ML-based DDoS detection using CICIDS 2017. - Ethical Approval Status: Approved - Ethical approval not required, as only publicly available datasets are used, and no human participants are involved.

Feedback and Reflection
- Feedback received from the supervisor or peers - Changes or adjustments made to the project based on feedback
Supervisor Feedback - Suggested narrowing the focus of the aim and objectives. - Emphasized the need to improve academic writing and citation consistency.

- Recommended focusing on practical implementation after the literature review.

Reflection

The feedback provided a clear direction, especially in refining the project scope and academic writing. The importance of a strong foundation in ML model evaluation and understanding the dataset structure was realized during early exploration.

Plans for the Next Phase

- *Planned activities for the next 6–8 weeks*
- *Anticipated challenges and potential solutions*

- Finalize and submit the complete literature review by mid-January.
- Begin implementation of the methodology, including:
 1. Preprocessing of the CICIDS 2017 dataset (normalization, handling imbalances).
 2. Applying Random Forest feature selection.
 3. Training baseline models (e.g., Random Forest, Logistic Regression).
- Start evaluating model performance with relevant metrics.
- Prepare for writing the Methodology and Results chapters.

Anticipated Challenges

Dataset imbalance may affect model performance.

Potential Solution: Apply oversampling or under sampling techniques such as SMOTE.

Hyperparameter tuning could require time and testing.

Solution: Use grid search and cross-validation for model optimization.

February Progress Review

Project Details	
Student Name:	Sanish Singh Maharjan
Student Number:	22085549
Student Email Address:	sanishs2.singhmaharjan@live.uwe.ac.uk
Date	February 2024
Supervisor Name	Jonathan White
Project Title:	Machine Learning for DDoS Attack Detection

Progress Update
- Provide a concise summary of the progress made since the December Review Dissertation – Findings, data collection (if applicable), and chapter drafts Technical – Literature review, Prototyping, testing and implementation progress
Dissertation: - The introduction chapter has been written. - Literature review drafting has begun but requires further development and synthesis. Technical: - Dataset (CICIDS 2017) preprocessing completed, including data cleaning, encoding, and feature selection using Random Forest importance. - Multiple machine learning models have been implemented and tested: Random Forest, Logistic Regression, Naïve Bayes, and XGBoost. - Model training and prediction times were measured to assess computational efficiency. - Initial evaluation metrics including confusion matrix analysis and feature importance visualizations have been generated.

Key Milestones Achieved

- *List of completed sections, modules, or objectives met*

- Dataset acquired and cleaned.
- Feature selection completed.
- Models implemented and evaluated: Random Forest, Logistic Regression, XGBoost
- Evaluation using confusion matrix, accuracy, precision, recall, and F1-score.
- Code prepared for supervisor review.

Challenges and Problem Solving

- **Major Challenges Encountered Since December** – Technical, theoretical, or logistical difficulties
- **How were these addressed?** – Solutions, adaptations and changes in approach

Challenge: Difficulty in selecting optimal features and handling class imbalance in the dataset.

Solution: Used Random Forest for feature importance and applied basic balancing techniques (e.g., sampling methods).

Challenge: Limited understanding of some evaluation metrics (e.g., ROC-AUC).

Solution: Referred to academic resources and online tutorials; will continue working on ROC curve implementation in the next phase.

Remaining Work and Completion Plan

- **Outstanding Tasks** – What remains to be completed
- **Estimated Timeline for Remaining Work** – Breakdown of tasks with estimated completion dates
- **Current Risks to Completion** – Any delays or obstacles foreseen

Outstanding Tasks:

- Complete and refine the literature review chapter.
- Finalize ROC under AUC evaluations.
- Analyze comparative performance of all models.
- Draft results discussion and conclusion chapters.
- Complete referencing and proofreading.

Estimated Timeline for Remaining Work:

Task	Target Completion
ROC Curve Evaluation	6 March 2025
Results Discussion & Evaluation	10 March 2025
Conclusion & Future Work	15 March 2025
Reference Management	20 March 2025
Document Refinement	10 April 2025

Final Review & Corrections	20 April 2025
Submission	30 April 2025

Student Reflection
• Confidence in Completion – <i>Do you feel on track for submission</i> • Support or Additional Resources Needed – <i>Any assistance required to help you complete the project?</i>
Feeling confident with the current progress and support. Technical skills have improved steadily. Library - Occasional guidance on interpreting evaluation metrics. - Continue using Studiosity and library sessions for writing clarity.

Critical Appraisal

This project was chosen due to my interest in DDoS attacks, as it was the first cyber-attack I knew before studying Cyber Security and Digital Forensics. My curiosity about how these attacks work, as well as how they can be detected and mitigated, led to the development of this project. During my research, I found that machine learning is a leading trend in detecting DDoS attacks. As a beginner in machine learning, I chose supervised learning as a foundation. This project explores how supervised machine learning models can effectively detect DDoS attacks while balancing accuracy, efficiency, and practical deployment.

The primary goal of this project was achieved through the development and comparison of four different models: XGBoost, Random Forest, Logistic Regression, and Naïve Bayes, using the CICIDS2017 dataset. Model evaluation was performed using various evaluation metrics and confusion matrices. Further comparisons were carried out using ROC curves, which revealed that XGBoost delivered exceptional results. XGBoost achieved 100% accuracy with a false alarm rate of just 0.01% and a training time of only 0.06 minutes, making it highly suitable for real-time detection systems.

The machine learning pipeline played a significant role in the project's success, as illustrated in Figure 2. Pre-processing the data was a time-consuming challenge, particularly handling missing values, infinite values, normalization, and label encoding. However, with the resources provided by my supervisor and past projects, these issues were resolved. Feature selection using Random Forest Feature Importance (RFFI) proved to be a pivotal step, reducing the feature set from 18 to 8 while maintaining model performance. This not only improved training efficiency but also confirmed that many original features were redundant. The application of stratified 5-fold cross-validation and sanity checks using label shuffling enhanced the reliability of the results, ensuring that model performance was not affected by data leakage or overfitting.

SWOT Analysis

For better reflection of this project comprehensively, a SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis is chosen:

Strengths

This project demonstrated high-performance models, particularly XGBoost, which exhibited superior accuracy in detecting DDoS attacks. Feature reduction techniques, such as Random Forest Importance, contributed to efficient model training and improved performance. Furthermore, robust validation methods lent credibility to the results. A practical focus was evident in the selection of evaluation metrics, including false alarm rate and training time, which are crucial for real-world applications and deployment feasibility.

Weaknesses

Despite its strengths, the study has some notable weaknesses. The research relies on a single, lab-generated dataset, which may not fully represent the diversity and unpredictability of real-world network traffic. The binary classification approach oversimplifies the detection task by failing to differentiate between various types of DDoS attacks. Additionally, the study does not evaluate model performance in real-time scenarios, limiting insights into practical deployment. Another significant drawback is the absence of explainability tools, reducing model

transparency and potentially hindering adoption in sensitive or regulated environments where interpretability is crucial.

Opportunities

Several promising opportunities exist to extend and improve this work. Expanding the study to multi-class classification would allow the detection of specific DDoS attack types rather than simply distinguishing between malicious and benign traffic. Applying these models to newer, more complex datasets—such as CICDDoS2019 or IoT-23—could enhance generalizability and robustness. Integrating explainability frameworks like SHAP (Shapley Additive Explanations) would provide insights into model decision-making, improving user trust and facilitating debugging. Additionally, evaluating performance under live traffic conditions would help validate the system's effectiveness in real-world applications.

Threats

Several external threats could impact the success and longevity of this research. The evolving nature of cyber threats means that new DDoS attack techniques could outpace static models, diminishing their effectiveness over time. Deploying these models in resource-constrained environments, such as IoT devices, poses technical challenges due to limitations in processing power and memory. Furthermore, there is a risk of overfitting, particularly if evaluation metrics are overly optimistic and fail to account for generalization, leading to unreliable performance in practical scenarios.

Challenges Encountered

During this project, several challenges were identified. Data processing proved to be more complex than anticipated, especially when handling anomalies such as missing or infinite values, which had to be carefully managed to ensure stability during training. Theoretical research was difficult initially, but attending library sessions helped me build a foundational understanding.

Lessons Learned

From this experience, several valuable lessons emerged. Technically, I learned that ensemble models like XGBoost and Random Forest are not only more accurate but also more robust under constrained conditions. Additionally, the importance of proper validation became clear as high accuracy can sometimes indicate overfitting, and I learned to verify models rigorously. Looking beyond accuracy and considering operational metrics, such as false alarm rate and training speed, proved crucial. Conceptually, this project strengthened my appreciation for scalability, model transparency, and the challenges of deploying AI in security-sensitive environments.

Future Improvements

Looking ahead, there are clear areas for future enhancement. Expanding the project to support multi-class classification would allow it to distinguish between various types of DDoS attacks. Real-time testing should be incorporated to measure inference speed and assess system performance under live traffic conditions. Testing on more diverse datasets would improve generalizability. Introducing explainability tools like SHAP would make model decisions more transparent, while adversarial testing could assess model resilience against obfuscation and evasion tactics.

Conclusion

In conclusion, this project successfully demonstrated the potential of machine learning, particularly ensemble models, for DDoS detection. It also highlighted the theoretical and technical considerations necessary for transitioning from proof of concept to real-world deployment. By addressing challenges related to dataset diversity, real-time performance, and model interpretability, future work can build upon this foundation to develop more adaptable, transparent, and resilient intrusion detection systems.

Confusion Matrix

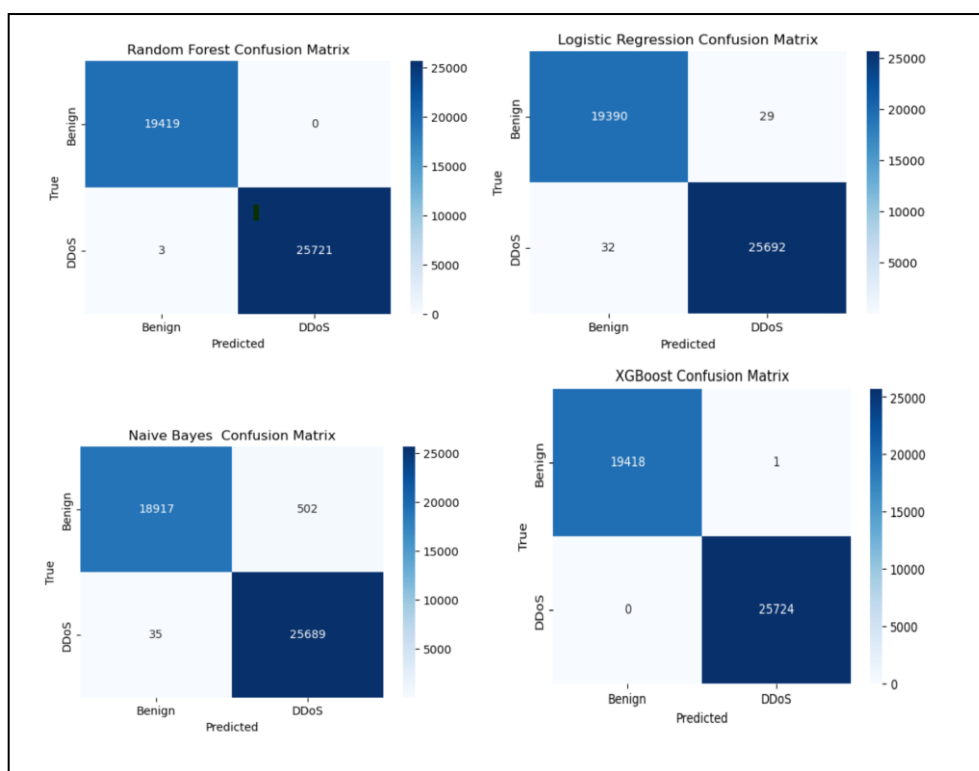


Figure 10: Confusion matrix on all feature set

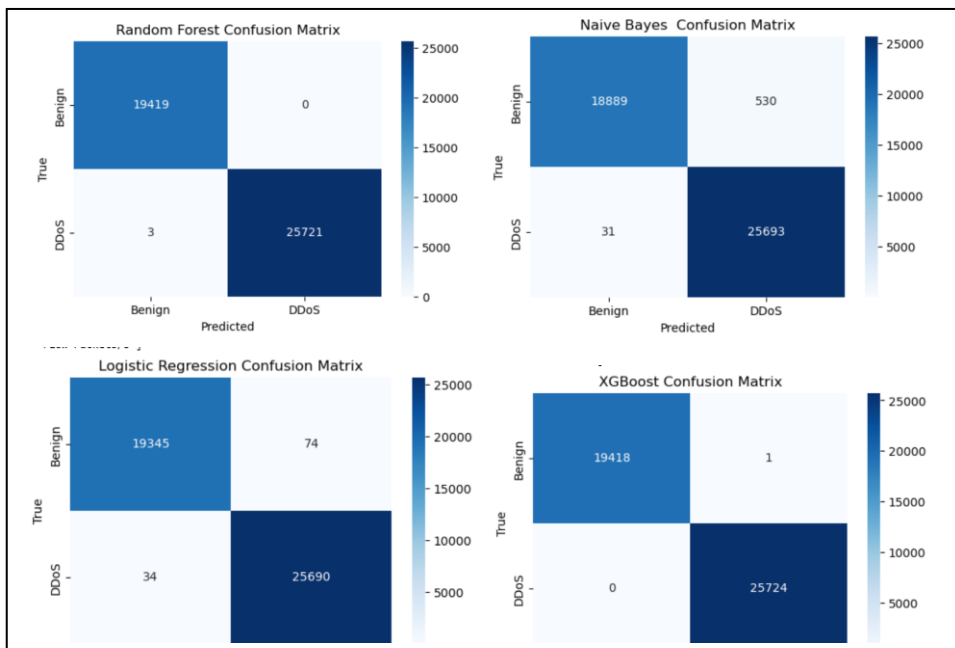


Figure 11: Confusion matrix on 53 feature set

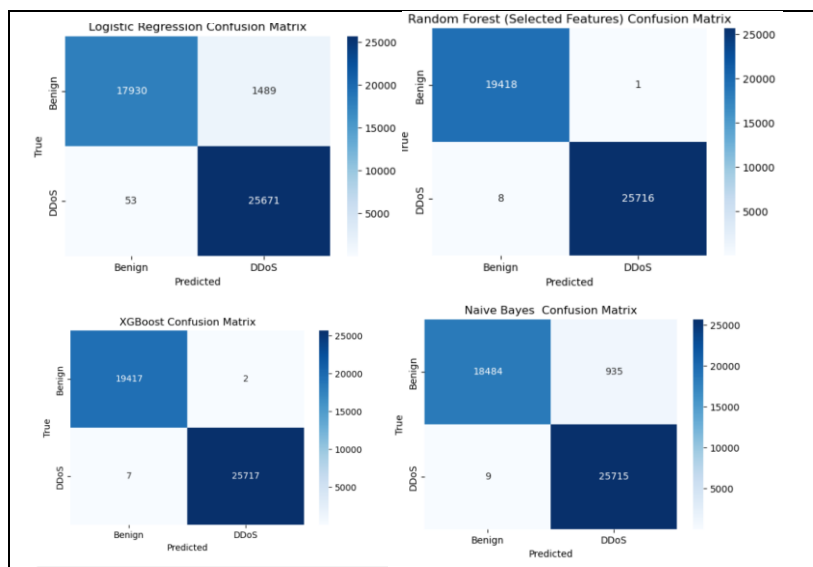


Figure 12: Confusion matrix on 13 feature set

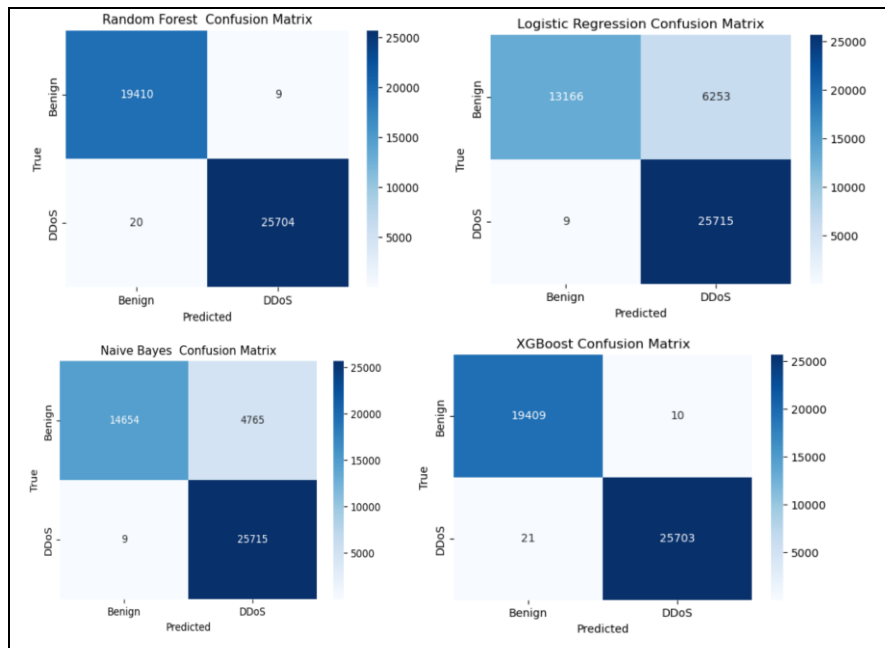


Figure 13: Confusion matrix on 8 feature set

Sanity check

8.2. Logistic Regression

```
]: import numpy as np

# Shuffle the labels
y_train_shuffled = np.random.permutation(y_train)

# Try cross-validation again
cv_scores_shuffled = cross_val_score(
    log_reg_model,
    X_train_selected,
    y_train_shuffled,
    cv=skf,
    scoring='accuracy',
    n_jobs=-1
)

print("\nCross-validation scores after shuffling labels for logistic Regression:", cv_scores_shuffled)
print(f"Mean CV accuracy (shuffled): {np.mean(cv_scores_shuffled) * 100:.2f}%")
```

Cross-validation scores after shuffling labels for logistic Regression: [0.56645622 0.56659467 0.56642853]
Mean CV accuracy (shuffled): 56.65%

Figure 14: Sanity check using Shuffling label for Logistic Regression on all feature set

8. Shuffle the Labels as a Sanity Test

8.1. Random Forest

```
import numpy as np
from sklearn.model_selection import StratifiedKFold, cross_validate
from sklearn.model_selection import cross_val_score

# Shuffle the labels
y_train_shuffled = np.random.permutation(y_train)

# Try cross-validation again
cv_scores_shuffled = cross_val_score(
    rf_model_selected,
    X_train_selected,
    y_train_shuffled,
    cv=skf,
    scoring='accuracy',
    n_jobs=-1
)

print("\nCross-validation scores after shuffling labels for Random Forest:", cv_scores_shuffled)
print(f"Mean CV accuracy (shuffled): {np.mean(cv_scores_shuffled) * 100:.2f}%")
```

Cross-validation scores after shuffling labels for Random Forest: [0.5215152 0.52187517 0.51968766 0.52222191 0.52366184]
Mean CV accuracy (shuffled): 52.18%

Figure 15: Sanitycheck using Shuffling label for Random Forest on all feature set

8.4. Naive Bayes

```
import numpy as np

# Shuffle the labels
y_train_shuffled = np.random.permutation(y_train)

# Try cross-validation again
cv_scores_shuffled = cross_val_score(
    naive_bayes_model,
    X_train_selected,
    y_train_shuffled,
    cv=skf,
    scoring='accuracy',
    n_jobs=-1
)

print("\nCross-validation scores after shuffling labels for Naive Bayes:", cv_scores_shuffled)
print(f"Mean CV accuracy (shuffled): {np.mean(cv_scores_shuffled) * 100:.2f}%")
```

Cross-validation scores after shuffling labels for Naive Bayes: [0.55158664 0.56468406 0.56543169 0.56525351 0.56209675]
Mean CV accuracy (shuffled): 56.18%

Figure 16: Sanitycheck using Shuffling label for NB on all feature set

8.3. XGBoost

```
] : import numpy as np

# Shuffle the Labels
y_train_shuffled = np.random.permutation(y_train)

# Try cross-validation again
cv_scores_shuffled = cross_val_score(
    xgb_model,
    X_train_selected,
    y_train_shuffled,
    cv=skf,
    scoring='accuracy',
    n_jobs=-1
)

print("\nCross-validation scores after shuffling labels for XGBoost:", cv_scores_shuffled)
print(f"Mean CV accuracy (shuffled): {np.mean(cv_scores_shuffled) * 100:.2f}%")
```

Cross-validation scores after shuffling labels for XGBoost: [0.55629396 0.55676469 0.55665393 0.55838618 0.55642013]
Mean CV accuracy (shuffled): 55.69%

Figure 17: Sanity check using Shuffling label for XGBoost on all feature set

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